

Automating SEM Microstructure Characterization for 3D Digital Microstructures

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Motivation

- Mechanical and thermal properties of cast irons are strongly governed by **graphite morphology**.
- SEM micrographs are routinely used for microstructural analysis, but **manual characterization is slow and subjective**.
- Current 2D metallography often relies on subjective visual comparison with DIN EN ISO 945 standard images [1].
- Quantitative descriptors such as **particle size, shape, and number density** are essential inputs for microstructure-based simulations.
- Automating microstructure characterization enables **reproducibility, scalability, and seamless digital microstructure generation**.

Problem Statement

- Given SEM micrographs of two cast iron grades:
 - **SiMo Ductile Iron** – Nodular (spheroidal) graphite morphology
 - **SiMoNi Compacted Graphite Iron** – Vermicular (worm-like) graphite morphology
- Automatically extract graphite particle statistics from 2D SEM images.
- Fit appropriate statistical distributions to the size data.
- Use the extracted statistics to drive an **existing MATLAB-based 3D two-phase microstructure generator**.
- Validate the pipeline against published literature benchmarks.
- Develop a simple **user interface** to integrate the complete workflow.

Spheroidal vs. Compacted Graphite

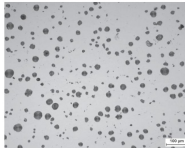


Figure 2.6 Cast iron with spheroidal graphite (optical micrograph, magnification 200x).

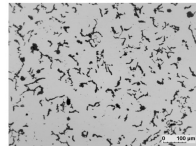


Figure 2.7 Cast iron with compacted graphite (optical micrograph, magnification 200x).

Spheroidal Graphite (SG)

Isolated nodules

Steel-like strength (~ 700 MPa)

Minimal notch effect

Compacted Graphite (CG)

Worm-like, intermediate

Good thermal conductivity

& mechanical strength

Reference: Velichko *et al.*, *Metallography, Microstructure, and Analysis* 7 (2008) 427–440 [1].

Material Overview: SiMo (SGI) vs. SiMoNi (CGI)

Paper: J. Hosdez *et al.*, *Int. J. Fatigue* 131 (2020) [6].

Chemical Composition & Microstructure:

- **SiMo (SGI):** 3.4% C, 3.89% Si, 0.57% Mo. Features isolated, spheroidal graphite nodules in a ferritic matrix.
- **SiMoNi (CGI):** 3.4% C, 5.0% Si, 0.8% Mo, 0.7% Ni. Features a complex 3D coral-like network of worm-shaped vermicular graphite.
- **Volume Fraction:** Despite entirely different shapes, both contain a similar total amount of graphite: $\approx 14.5\%$ (SGI) and 15.7% (CGI).

3D X-ray Microtomography statistics:

- SGI: mean sphericity 0.93 ± 0.09 ; mean 3D diameter $26 \pm 7 \mu\text{m}$.
- CGI: mean sphericity 0.72 ± 0.23 ; vermicule mean Feret length $64 \pm 35 \mu\text{m}$.

3D Graphite Morphology (Hosdez et al.)

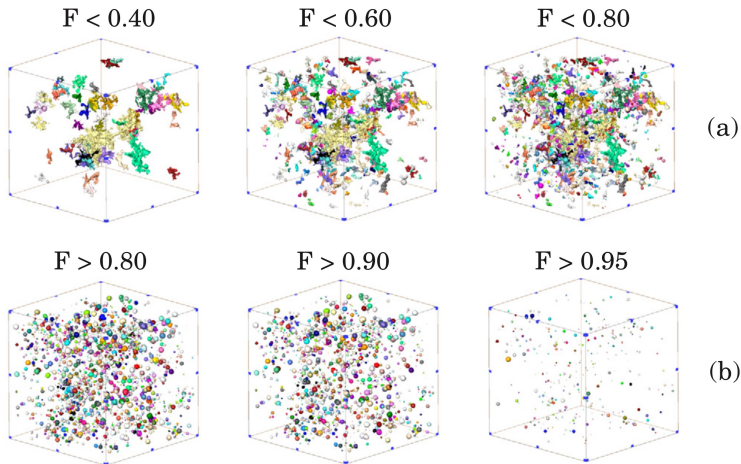


Fig. 6. Rendering of graphite particles at different levels of sphericity in CGI: vermicules (a) and nodules (b).

Hosdez et al. [6]. Sphericity: $F = 6\sqrt{\pi} \frac{\text{Volume}}{\sqrt{\text{Area}}^3}$.

Effect of Cooling Rate on Microstructure [7]

Paper: M. Górny *et al.*, *Metals* 11 (2021) 1634 – EN-GJS-SiMo45-6.

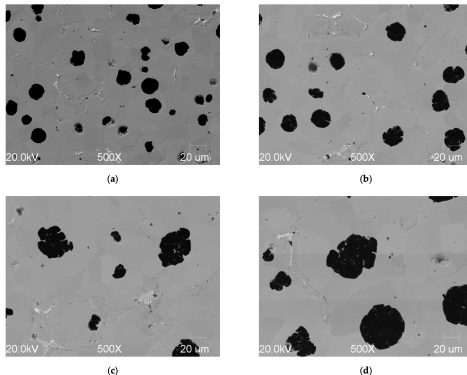


Figure 5. SEM microstructure of Si-Mo ductile iron in castings with different wall thicknesses: (a) 3 mm; (b) 5 mm; (c) 13 mm; and (d) 25 mm. Nital-etched samples.

Microstructure of Si-Mo ductile iron for wall thicknesses: (a) 3 mm, (b) 5 mm, (c) 13 mm, (d) 25 mm. Faster cooling (thinner wall) produces finer, denser nodules.

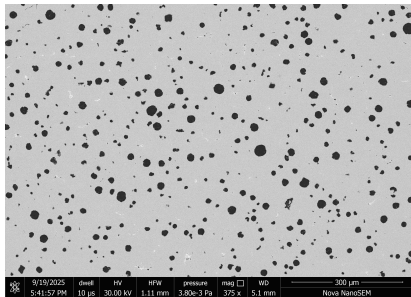
Quantitative Benchmarks from Literature [7]

Wall	Nodule Density	Mean Diameter	Nodularity
3 mm (fastest)	$608 \pm 35 \text{ mm}^{-2}$	$15.32 \pm 0.25 \mu\text{m}$	91%
5 mm	$411 \pm 28 \text{ mm}^{-2}$	$19.17 \pm 0.28 \mu\text{m}$	90%
13 mm	$201 \pm 13 \text{ mm}^{-2}$	$26.27 \pm 1.11 \mu\text{m}$	87%
25 mm (slowest)	$168 \pm 15 \text{ mm}^{-2}$	$27.19 \pm 0.81 \mu\text{m}$	85%

Key trend: Faster cooling $\rightarrow$ higher density, smaller, more uniformly spheroidal nodules.

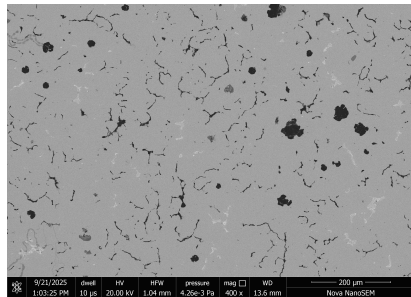
Key finding: At low cooling rates, size distribution transitions from unimodal to bimodal due to secondary nucleation.

Dataset Description



SiMo Ductile Iron

Nodular graphite



SiMoNi Compacted Graphite Iron

Vermicular graphite

- **SiMo:** 8 TIFF images; resolution $\approx 2207 \times 3072$ pixels; varying scale bars (100–400 μm).
- **SiMoNi:** 8 images; flake-like and vermicular graphite; acquired at different magnifications.
- Dark regions correspond to the graphite phase; bright regions represent the iron matrix.

Extracted Graphite Statistics for 3D Reconstruction

- **Phase Fraction**

$$f_g = \frac{A_{\text{graphite}}}{A_{\text{total}}} \Rightarrow f_g \approx V_g$$

- **Particle Size (Equivalent Diameter)**

$$d_{\text{eq}} = 2\sqrt{\frac{A}{\pi}}$$

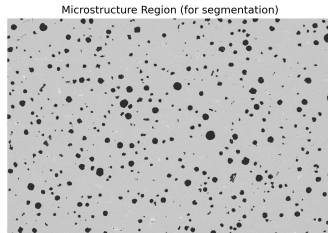
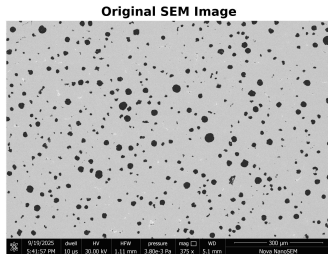
- **Particle Shape**

$$\text{Aspect Ratio} = \frac{l_{\text{major}}}{l_{\text{minor}}}, \quad \text{Circularity} = \frac{4\pi A}{P^2}$$

- **Number Density**

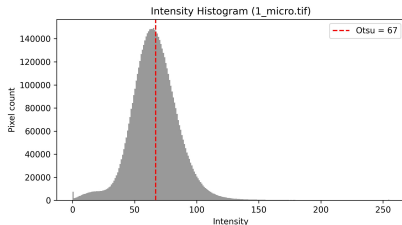
$$N_A = \frac{N_{\text{particles}}}{A_{\text{image}}}$$

Metadata Strip Separation

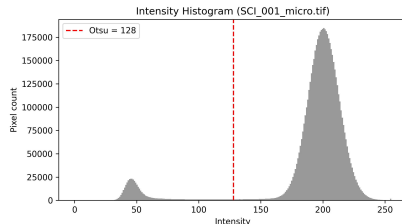


- SEM images contain metadata strip at bottom (~7.3% of height)
- Separation required before segmentation
- Metadata preserved separately for record-keeping

Intensity Distribution Analysis



1_micro.tif (mean = 66.0)



SCI_001_micro.tif (mean = 129.0)

Two distinct cases identified:

- **Case 1 (1/8 images):** Dark background (mean ≈ 66) — challenging separation
- **Case 2 (7/8 images):** Bright background (mean ≈ 129) — clear bimodal distribution
- Case 1: Requires advanced multi-step processing (But was eventually discarded due to poor quality and outlier statistics)
- Case 2: Simple Otsu thresholding works well

Scale Calibration Pipeline

Workflow – Automated $\mu\text{m}/\text{pixel}$ extraction from SEM metadata strip:

① Scale Bar Detection

- Morphological analysis of the metadata strip at image bottom ($\approx 7.3\%$ of height).
- Handles short ($100\ \mu\text{m}$) and long ($400\ \mu\text{m}$) bars; aspect ratio constraint: $\text{width} > 2 \times \text{height}$.

② OCR Text Extraction

- Multiple Tesseract attempts (PSM 7, PSM 8; inverted and original).
- Sanity checking: $1\text{--}1000\ \mu\text{m}$ range; majority voting across candidates.

③ Calibration Calculation

$$\mu\text{m}/\text{pixel} = \frac{\text{scale value } (\mu\text{m})}{\text{bar length (pixels)}}$$

Success Rate: 8/8 SiMo images (100%); handles $100\text{--}400\ \mu\text{m}$ scale bars.

Scale Calibration Results: All 8 SiMo Images

Image	Scale (μm)	Pixels	$\mu\text{m}/\text{pixel}$
1	400	744	0.53763
SCI_001	300	836	0.35885
SCI_002	300	836	0.35885
SCI_003	300	836	0.35885
SCI_004	100	521	0.19194
SCI_005	100	521	0.19194
SCI_006	100	521	0.19194
SCI	200	299	0.66890

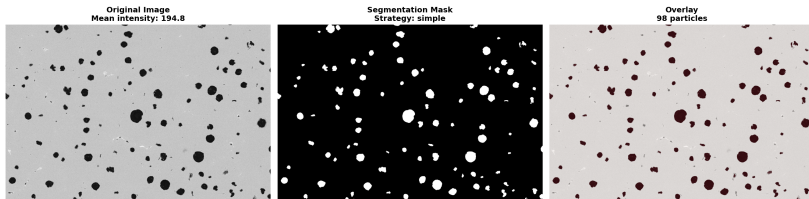
Segmentation Refinement

Additional filters added (physically motivated):

- `binary_fill_holes`: fills fragmented nodule borders
- Minimum diameter filter: $d_{\min} = 5.0 \mu\text{m}$
- Minimum circularity: $C_{\min} = 0.2$ (removes elongated artifacts)
- `min_area_px` derived from scale, consistent across magnifications

Image	Before Refinement	After Refinement
SCI_001	516	391
SCI_002	472	364
SCI_003	421	318
SCI_004	164	100
SCI_005	147	101
SCI_006	173	98
SCI	371	332
Total	2264	1704

Refined Segmentation Demo (SCI_006)



Example of refined segmentation after morphological cleanup and size/circularity filtering. Detected nodules shown in overlay; grey regions are rejected candidates.

Material-Specific Adaptation: SiMoNi

After validating on SiMo, the pipeline was extended to **SiMoNi Compacted Graphite Iron**.

Challenges:

- Graphite morphology differs significantly – vermicular / flake-like structures.
- Circularity-based filtering used for nodular graphite rejects valid particles.
- Higher noise levels in some SiMoNi images.

Adjusted Parameters:

- Minimum graphite diameter reduced to:

$$d_{min} = 0.5 \mu\text{m}$$

- Circularity filtering removed:

$$C_{min} = 0$$

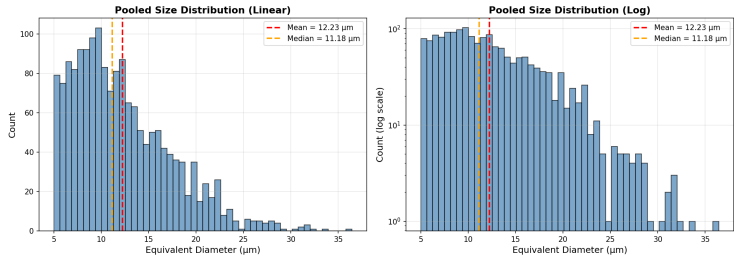
Vermicular/flake graphite typically has low circularity; a high circularity cutoff can incorrectly remove valid particles.

Batch Processing Results: SiMo (7 Images)

Image	Strategy	Particles	f_g	Mean Int.
SCI_001	Simple	391	0.0669	190.9
SCI_002	Simple	364	0.0594	191.6
SCI_003	Simple	318	0.0617	191.9
SCI_004	Simple	100	0.0751	190.3
SCI_005	Simple	101	0.0618	192.6
SCI_006	Simple	98	0.0516	194.8
SCI	Simple	332	0.0636	192.8
Total	—	1704	0.0619*	—

* Mean phase fraction across all images. (Dark-background outlier image “1” excluded.)

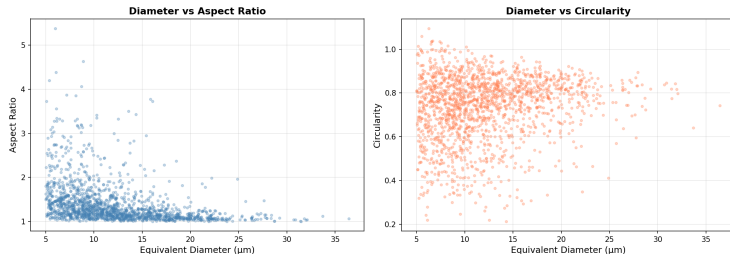
Pooled Size Distribution: SiMo



Equivalent Diameter Statistics (refined dataset):

- Mean $\pm$ Std: **12.23 $\pm$ 5.21 μm**
- Median: **11.18 μm**
- Range: 5.01 – 36.44 μm
- Total particles: **1704**

Shape Characteristics: SiMo



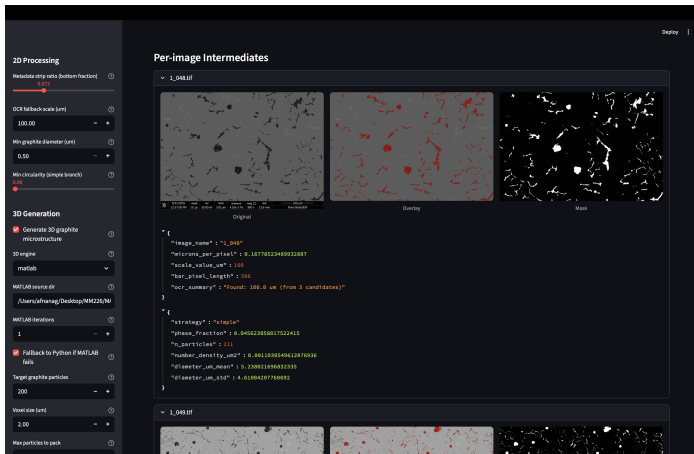
Morphological Metrics (refined):

- Aspect Ratio: 1.397 ± 0.473 (nodules are nearly equiaxed)
- Circularity: 0.735 ± 0.148
- Consistent with *Slightly Irregular Spheroidal Graphite* (ISO Form V) [8]

Summary Statistics for 3D Reconstruction: SiMo

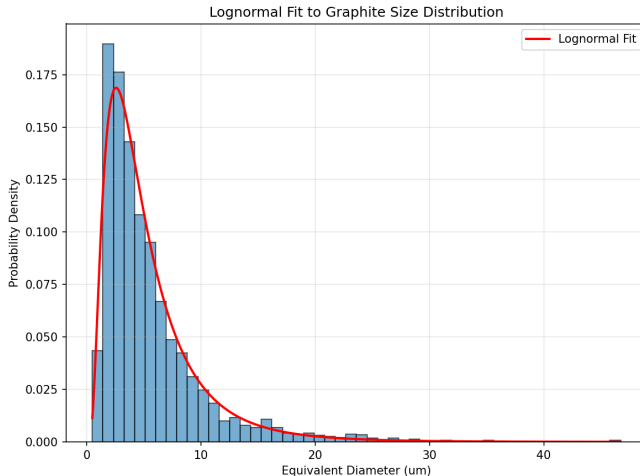
Parameter	Value
Phase Fraction ($f_g \approx V_g$)	0.0619 ± 0.0073
Mean Equivalent Diameter	$12.23 \pm 5.21 \text{ }\mu\text{m}$
Median Diameter	$11.18 \text{ }\mu\text{m}$
Diameter Range	$5.01\text{--}36.44 \text{ }\mu\text{m}$
Mean Aspect Ratio	1.397 ± 0.473
Mean Circularity	0.735 ± 0.148
Number Density	$4.42 \times 10^{-4} \text{ }\mu\text{m}^{-2}$
Total Particles Analyzed	1704
Number of Images	7

SiMoNi: Example Segmentation Output



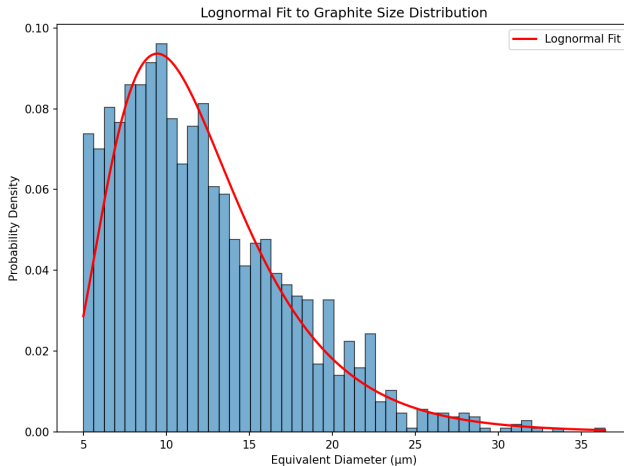
The UI provides intermediate outputs including the original SEM image, detected graphite regions, binary mask, and computed statistics.

SiMoNi: Graphite Size Distribution



Equivalent graphite diameters extracted from segmented particles fitted using a lognormal distribution, consistent with commonly observed particle size statistics.

Lognormal Distribution Fit: SiMo (Refined)



Fitted Parameters:

- $\mu = 2.4181, \quad \sigma = 0.4134$
- $\ln(D) \sim \mathcal{N}(\mu, \sigma^2)$

2D-to-3D Reconstruction Strategy

Objective: Generate a synthetic 3D graphite morphology from pooled 2D statistics.

MATLAB-based Graphite-Only 3D Reconstruction

(`graphite_only_main.m`):

- Explicit phase: Graphite nodules modeled as **ellipsoids**
- Matrix: implicit background (not packed as particles)
- Size model: Lognormal diameter distribution (fitted from 2D data)
- Orientation: random
- Multiple stochastic realizations (`n_iter`)

Representative Volume Estimate:

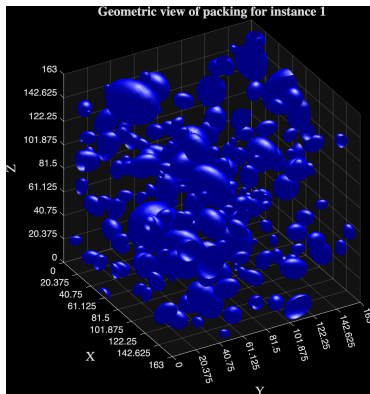
$$L \approx \left(\frac{N_{\text{target}} V_{\text{particle,mean}}}{f_g} \right)^{1/3}, \quad N_{\text{target}} \approx 200$$

where $V_{\text{particle,mean}}$ is estimated from mean diameter and aspect ratio.

Input Parameters: SiMo 3D Reconstruction

Pooled inputs used:

- Graphite volume fraction: $f_g = 0.0619$
- Mean diameter: $\bar{D} = 12.23 \mu\text{m}$, Std: $\sigma_D = 5.21 \mu\text{m}$
- Mean aspect ratio: $AR = 1.397$
- Lognormal fit: $\mu = 2.4181$, $\sigma = 0.4134$



Example 3D packing output: stochastic realization of graphite ellipsoids in the synthetic volume.

End-to-End Workflow via UI

System provides a single interface to control the entire workflow:

- ➊ **Input:** Upload SEM micrograph + select material type (SiMo / SiMoNi)
- ➋ **Preprocess:** Automatic scale extraction via OCR, metadata strip removal
- ➌ **Segment:** Material-adaptive graphite segmentation pipeline
- ➍ **Analyze:** Extract morphological statistics (size, shape, density, phase fraction)
- ➎ **Fit:** Statistical distribution fitting (Lognormal / Gamma); model comparison
- ➏ **Generate:** MATLAB backend for 3D ellipsoid packing
- ➐ **Export:** Downloadable particle CSV, summary statistics, 3D visualization

Key Features:

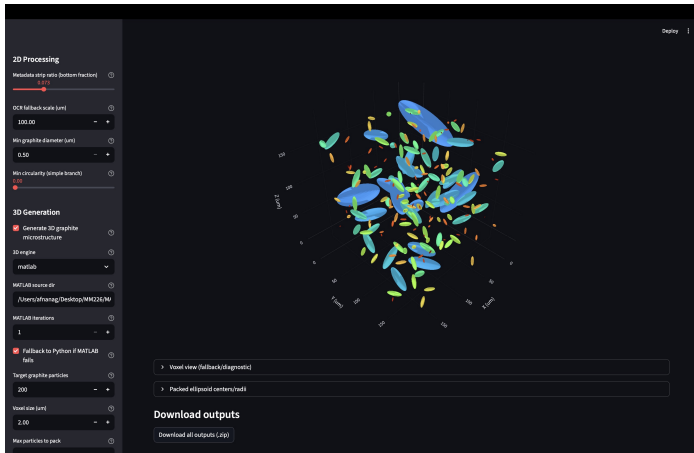
- Adjustable segmentation parameters via sliders
- Python fallback visualization if MATLAB unavailable

Streamlit-Based Integrated UI (SiMo)



The UI integrates the entire workflow from 2D SEM image processing to 3D graphite microstructure reconstruction using the MATLAB backend [10, 9].

Streamlit-Based Integrated UI (SiMoNi)



The extracted statistics are directly passed to the MATLAB ellipsoid-packing engine to generate synthetic 3D graphite microstructures.

Validation Setup: Górný et al. (2021) 3 mm Wall [7]

Objective: Run the full 2D→3D pipeline on the 3 mm wall microstructure image from Górný et al. and compare against reported statistics.

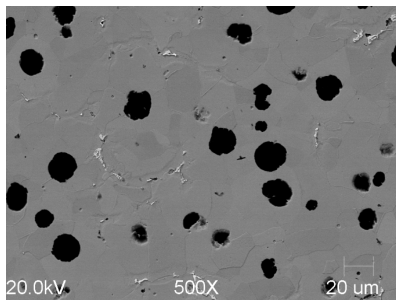
Why the 3 mm wall?

- Represents the **fastest cooling rate** – most microstructurally refined.
- Provides the **highest nodule density** (608 ± 35 nodules/mm²) – most demanding test.
- Tight unimodal size distribution is well-suited to a LogNormal fitting approach.

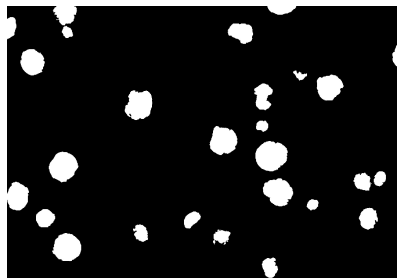
Ground-truth targets (from paper):

- Nodule density: 608 ± 35 nodules/mm²
- Mean diameter: 15.32 ± 0.25 μ m
- Nodularity: 91%, Ferrite fraction: 86%

Validation: Input Image & Segmentation Mask [7]



Input SEM image: 3 mm wall microstructure (Górny et al., Fig. 5a)



Model segmentation mask:
detected graphite nodules

Number of particles detected in the image: 27

Validation: Model Output & Lognormal Fit

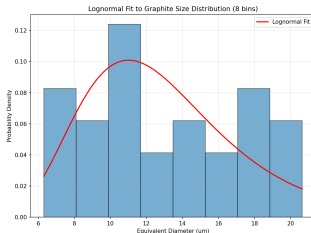
Graphite-Only Run: Fitted Parameters

Geometry:

- $V_f^{\text{graphite}} = 0.082907$
- $\bar{d} = 13.08 \mu\text{m}$
- $\sigma_d = 4.57 \mu\text{m}$
- $\bar{\alpha}$ (aspect ratio) = 1.436

Distribution Fit (LogNormal):

- $\mu_{\log} = 2.5136$
- $\sigma_{\log} = 0.3392$
- Mesh length = $179 \mu\text{m}$



LogNormal fit overlaid on the detected particle size distribution histogram.

Validation: Model vs. Paper [7]

Paper (Ground Truth – 3 mm Wall)

Nodule density: 608 ± 35 nodules/mm²
 Mean diameter: 15.32 ± 0.25 μ m
 Nodularity: 91%
 Ferrite fraction: 86%
 Distribution: unimodal

Model Output

Nodule density: 560 nodules/mm²
 Mean diameter: 13.08 μ m
 Std. deviation: 4.57 μ m
 $V_f = 0.0829$, $\bar{\alpha} = 1.436$
 LogNormal ($\mu = 2.514$, $\sigma = 0.339$)

Key Observations:

- Nodule density: 560 vs. 608 ± 35 nodules/mm² ($\approx 7.9\%$ underestimate).
- Mean diameter underestimated by $\approx 14.6\%$ (due to touching nodule merging in dense packing).
- LogNormal fit consistent with the expected unimodal distribution at fast cooling rates.

Conclusion

Achievements:

- Developed a **fully automated** pipeline for SEM microstructure characterization:
 - Scale calibration via OCR (100% success, 8/8 images)
 - Material-adaptive segmentation (SiMo nodular + SiMoNi vermicular)
- Extracted **quantitative graphite statistics** in physical units (μm):
 - Phase fraction, size distribution, shape descriptors, number density
- Implemented **MATLAB-based 3D graphite reconstruction** with Python-MATLAB Engine integration.
- **Validated** against Górný et al. (2021): nodule density within $\approx 8\%$, mean diameter within $\approx 15\%$.
- Delivered an **end-to-end Streamlit UI** integrating all steps.

References I

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